Computer Vision

Image Formation

Image Formation

Human Driver	Autonomous Vehicle

Image Formation

- how images are formed
- How cameras capture images
- How images are represented in computers

What we see

What a computer see?

What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer see

- Computer vision is **Making sense of these numbers**

What we see

Goal:

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

How this is generated?

Image Formation: Digital Image

- An Image is a **two dimensional function f(x, y),** where x and y are the **spatial (plane) coordinates** and the **amplitude of f at any pair of coordinates (x, y)** is called the **intensity of the image** at that level.

- **If x , y and the amplitude values of f are finite and discrete quantities, we call the image a digital image.** A digital image is composed of a finite number of elements called pixels, each of which has a particular location and value.

Image Formation

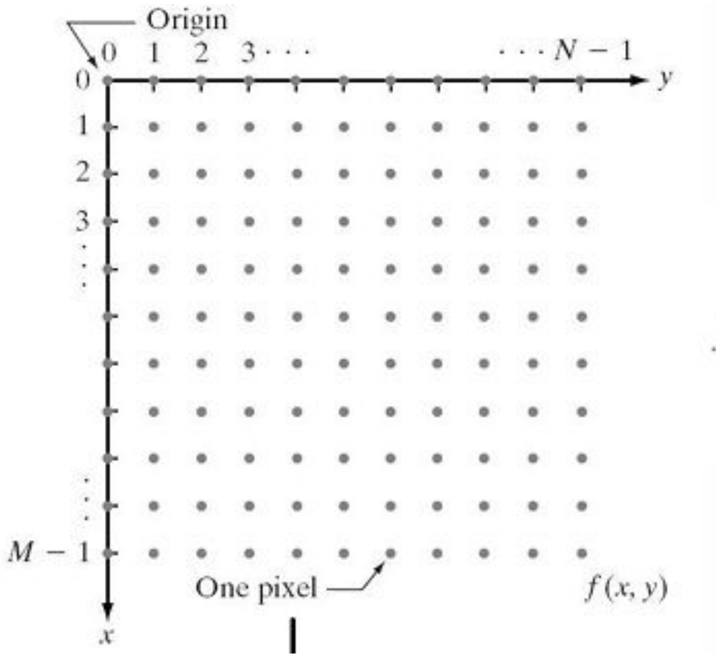
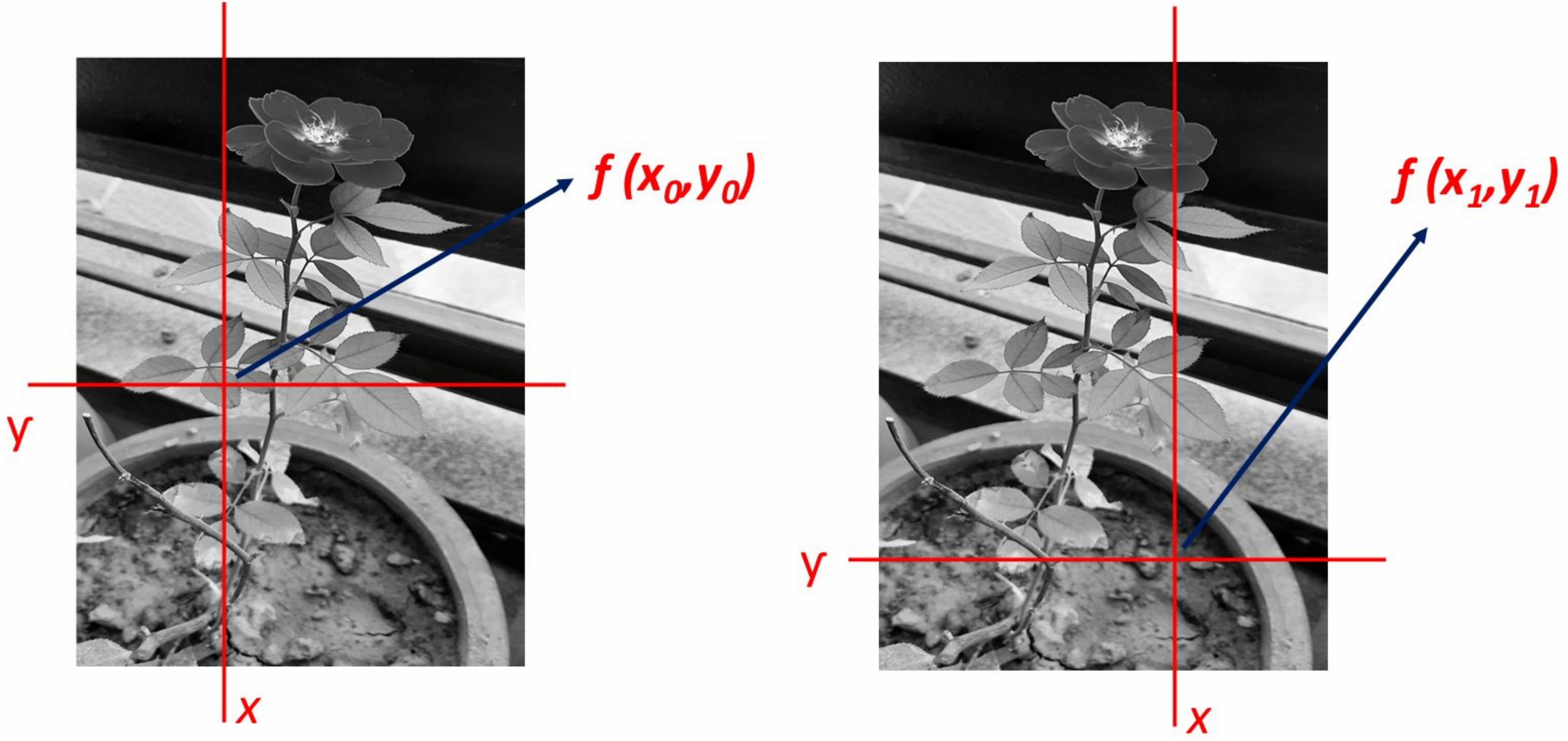


Image Formation



Activity: Identify 5 objects in the image

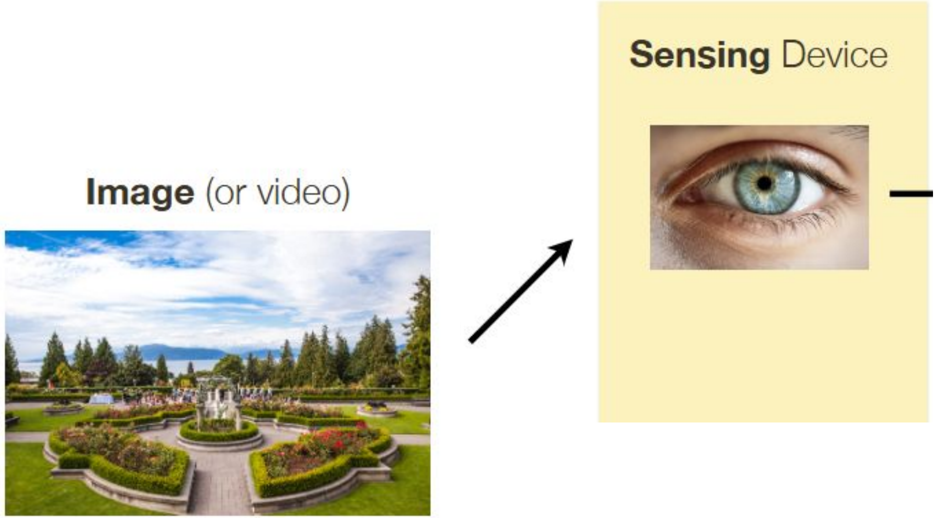


Image (or video)
Sensing Device

Image (or video)

Sensing Device

Interpreting Device

Image Credit: https://www.flickr.com/photos/flamephoenix1991/8

Image (or video)

Sensing Device

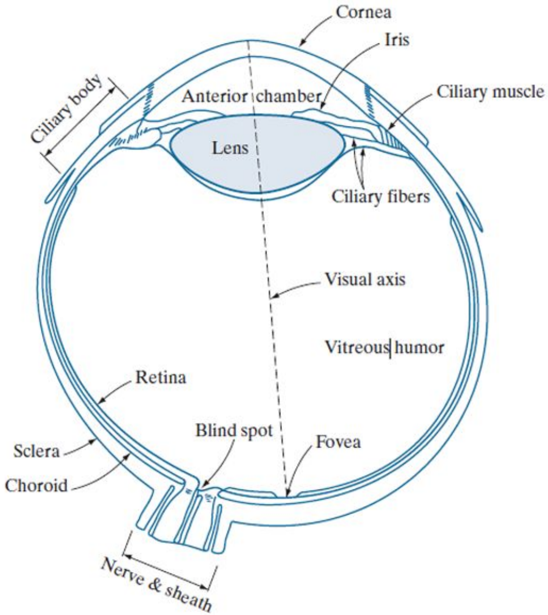
Interpreting Device

Image Credit: https://www.flickr.com/photos/flamephoenix1991/8376271918

Interpretation

blue sky,
trees,
fountains,

Image Formation in the Human Eye



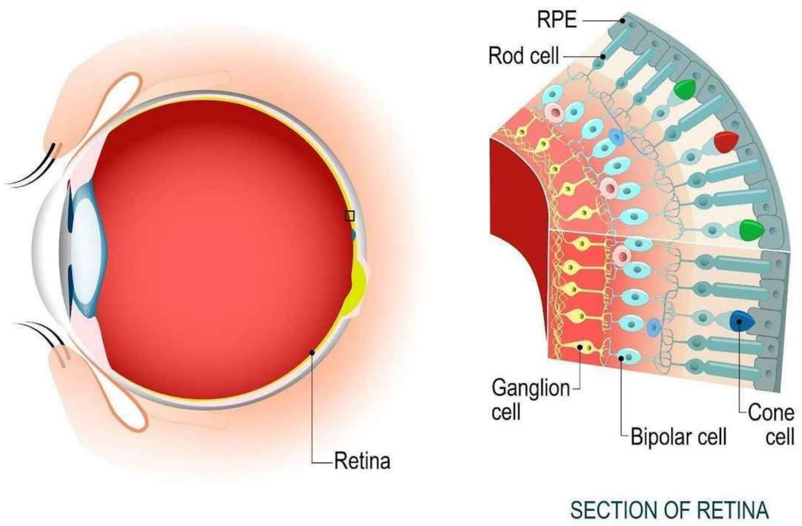


Image Formation in the Human Eye

- There are two types of receptors: cones and rods
- **Cones**
  - Between 6 and 7 million cones in each eye
  - Located in the central portion of the retina (fovea)
  - **Highly sensitive to color**
- Humans can resolve fine details because each cone is connected to its own nerve end
- Muscles rotate the eye until the image of a region of interest falls on the fovea
- **Cone vision is called photopic or bright-light vision**

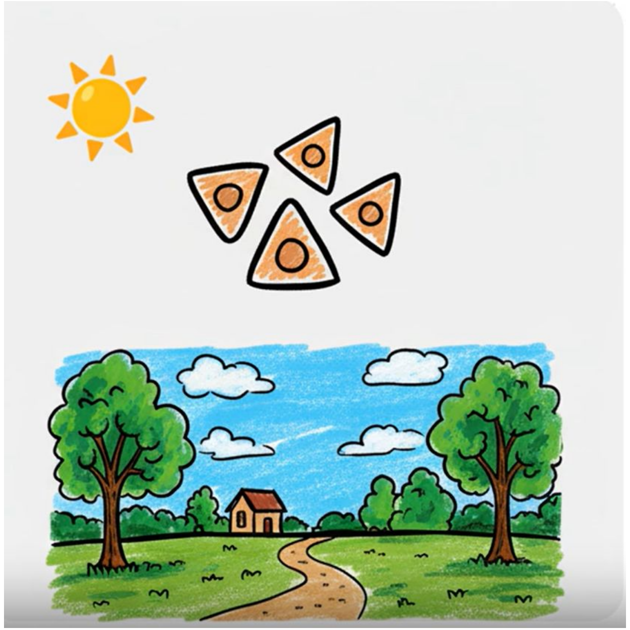
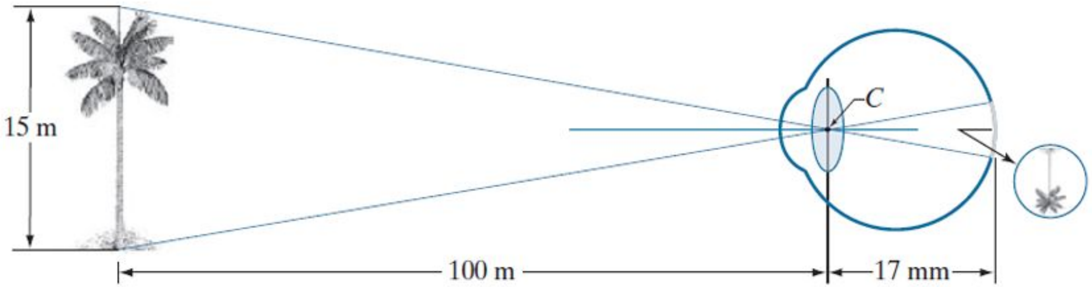


Image Formation in the Human Eye

- There are two types of receptors: cones and rods
- **Rods**
  - 75 to 150 million, distributed over the retina
  - Larger area of distribution and several rods being connected to a single nerve ending reduces the amount of detail discernible
  - Rods capture **an overall image of the field of view**.
- Rods are sensitive to low level illumination
  - Eg: Objects that appear brightly colored in daylight appear as colorless forms in the moonlight because only the rods are stimulated.
- **Rod vision is known as scotopic or dim-light vision**

How the eye forms an Image



- Perception takes place by the relative excitation of light receptors, which transform radiant energy into electrical impulses that are decoded by the brain.
  - **Light reflects from objects**
  - **Light enters the eye**
  - **Lens focuses light**
  - **Image forms in retina**

How can we build systems that allow computers to acquire visual information, much like an eye

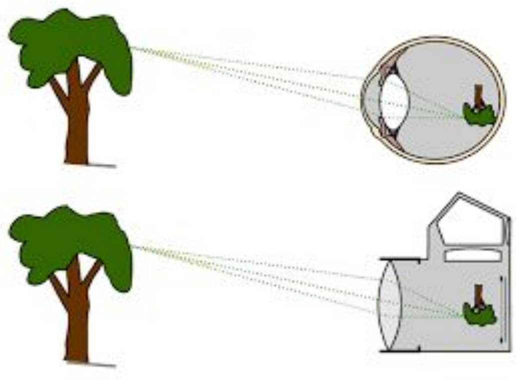
How can we build systems that allow computers to acquire visual information, much like an eye

First the system needs a device to capture visual data

Camera

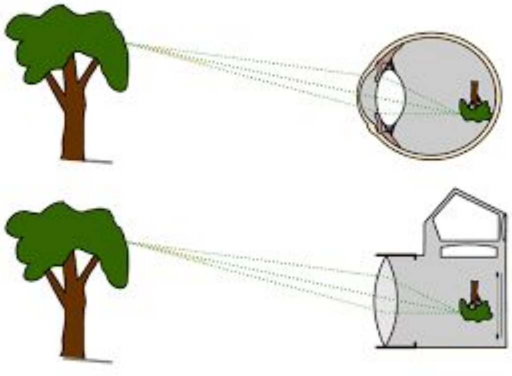
Eye vs Camera

Human Eye	Camera
Lens	
Retina	
Brain	



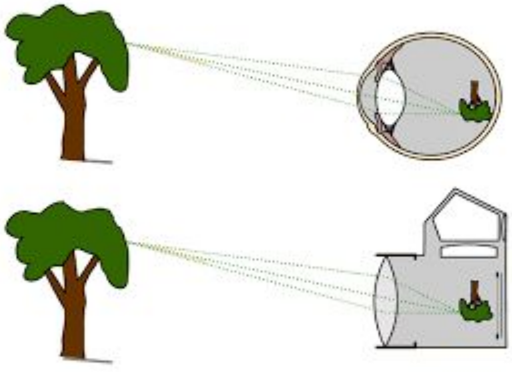
Eye vs Camera

Human Eye	Camera
Lens	Camera Lens
Retina	
Brain	



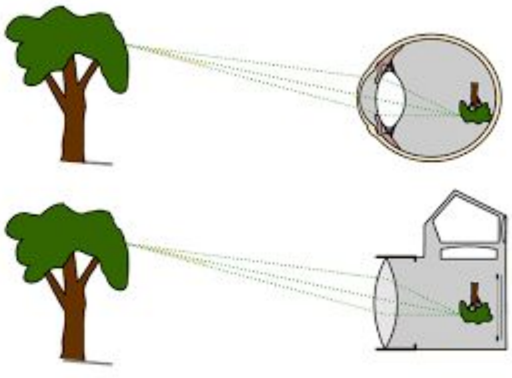
Eye vs Camera

Human Eye	Camera
Lens	Camera Lens
Retina	Image sensor
Brain	



Eye vs Camera

Human Eye	Camera
Lens	Camera Lens
Retina	Image sensor
Brain	Computer



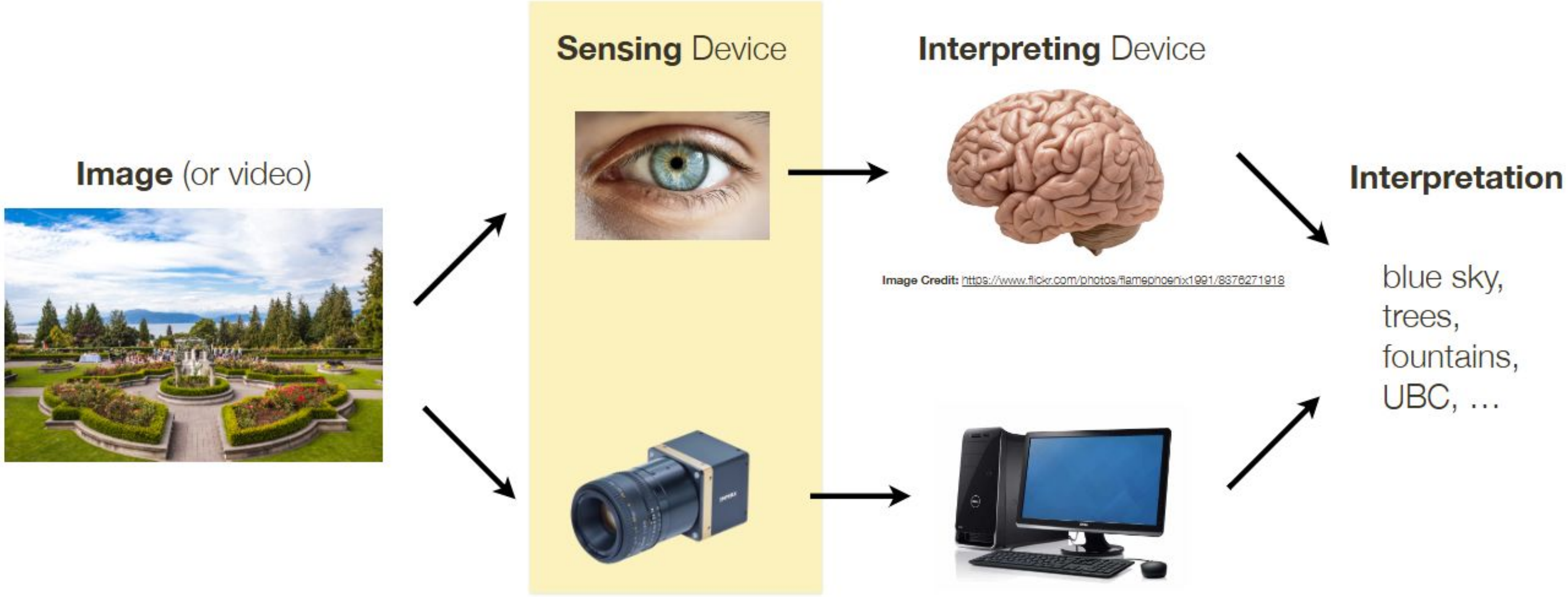
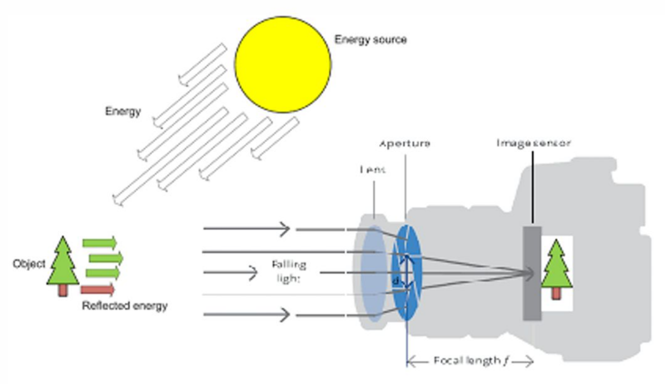
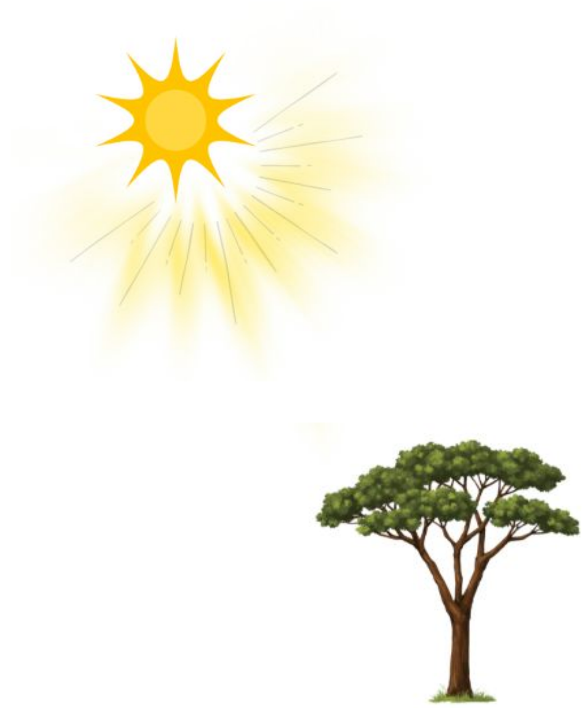


Image (or video)
Sensing Device
Interpreting Device
Image Credit: https://www.flickr.com/photos/flamephoenix1991/8376271918
Interpretation
blue sky,
trees,
fountains,
UBC, …

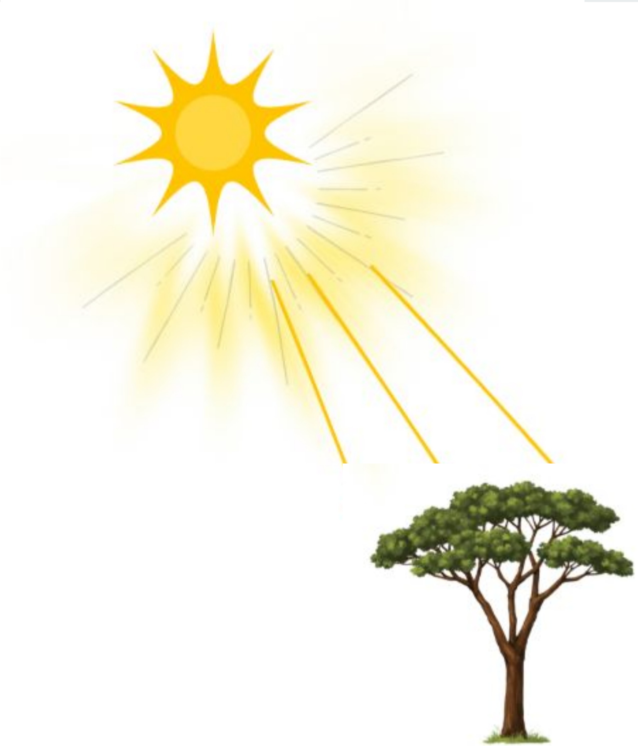
Image sensing and acquisition

- In image acquisition there are three components
  - Illumination
  - Optical system (lens system)
  - Sensor system



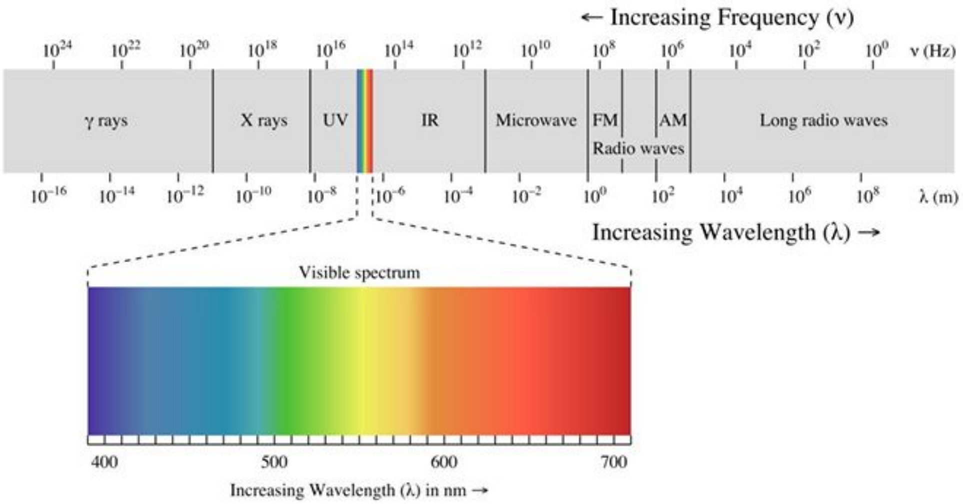


Step 1: Light source illuminate the scene



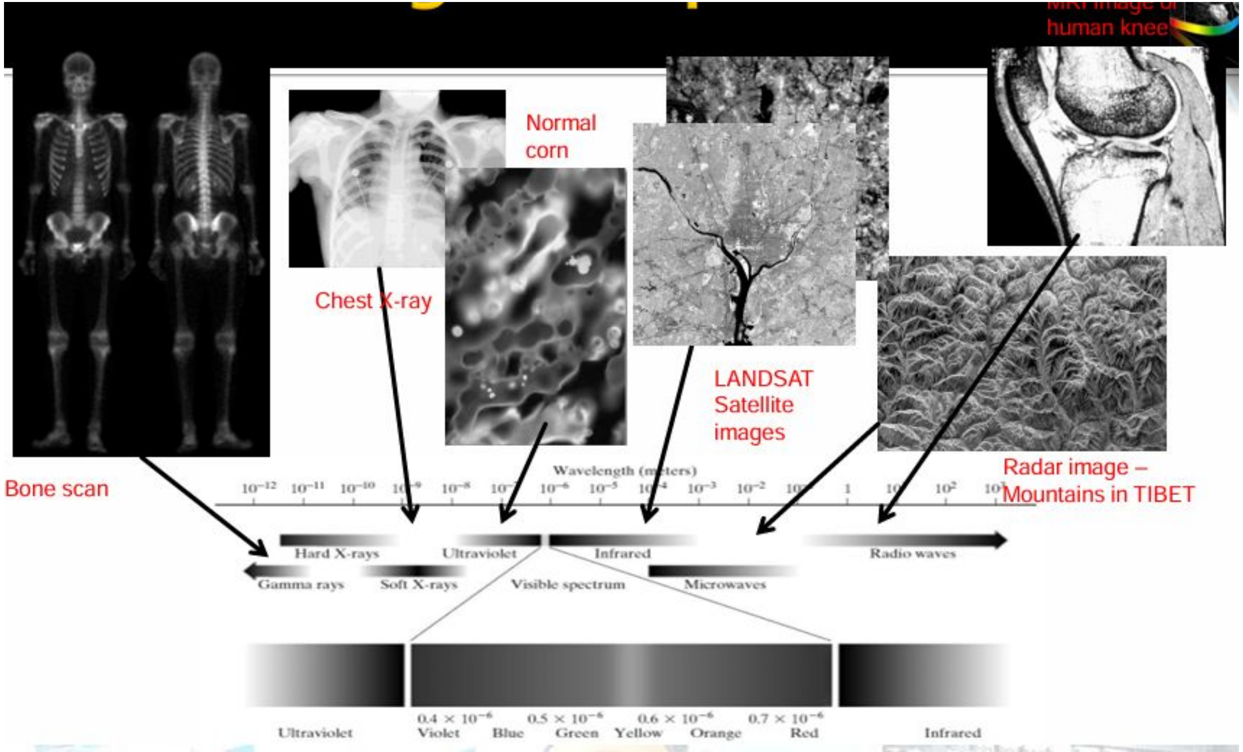
Step 1: Light source illuminate the scene and hit the objects in the environment

The Electromagnetic Spectrum



Primary energy source for images used in many applications. EMS can be expressed in terms of wavelength (λ) or frequency (v)

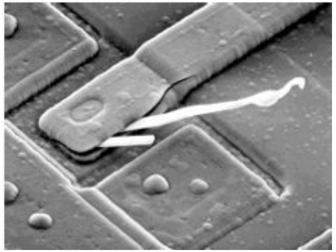
The Electromagnetic Spectrum



- Imaging is based mainly on energy radiated by electromagnetic waves.

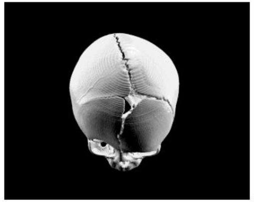
Other Imaging Methods

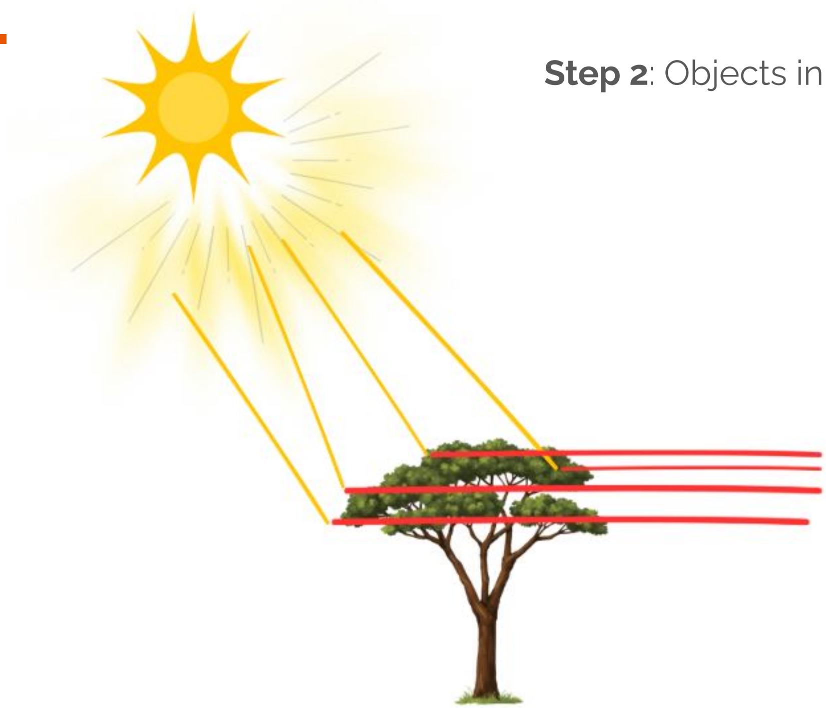
- Acoustic - sound waves used in geology, ultrasound imaging etc.



2500 x SEM image of a damaged integrated circuit

- Electron Microscopy
    - TEM - Transmission Electron Microscope
    - SEM - Scanning Electron Microscope
- Synthetic Images
    - Intersection between image processing and graphics.
    - Used in 3D visualization systems (For medical training etc)

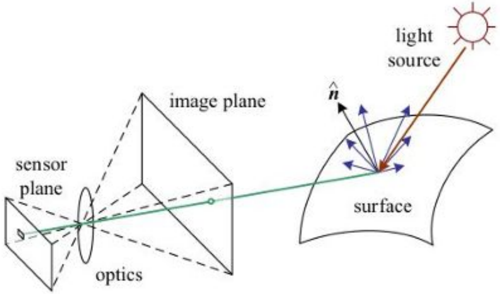




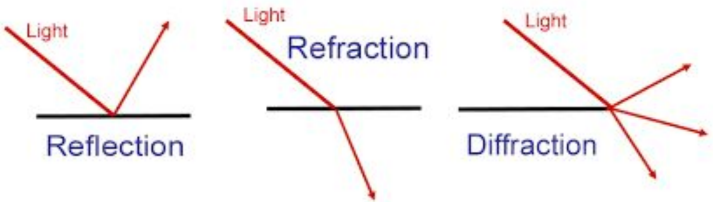
Step 2: Objects in the scene reflect light

All starts with light ...

- To being able to see any 3D scene, our eyes or a camera needs to capture the light radiation reflected from scene surfaces.
- To produce an image the scene must be illuminated with one or more light sources.



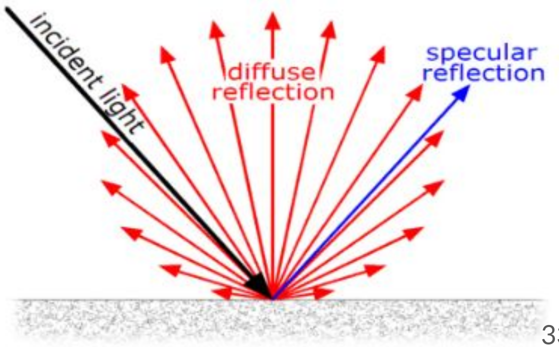
- Light travels as **electromagnetic waves.**
- It behave like a wave
    - **Refraction**
    - **Diffraction**
    - Have a wavelength and an amplitude



Reflection

- Incident light is reflected in two main forms
    - **Diffuse reflection** - light scattered isotropically in all direction (shows true color of the object) - show the true color of the pbject
    - **Specular reflection** - incident light reflected in a specific direction (mirror like effect) - shows the color of the light Source (usually white)

- Most materials exhibit a mixture of diffuse and specular reflections.



Light Quantities

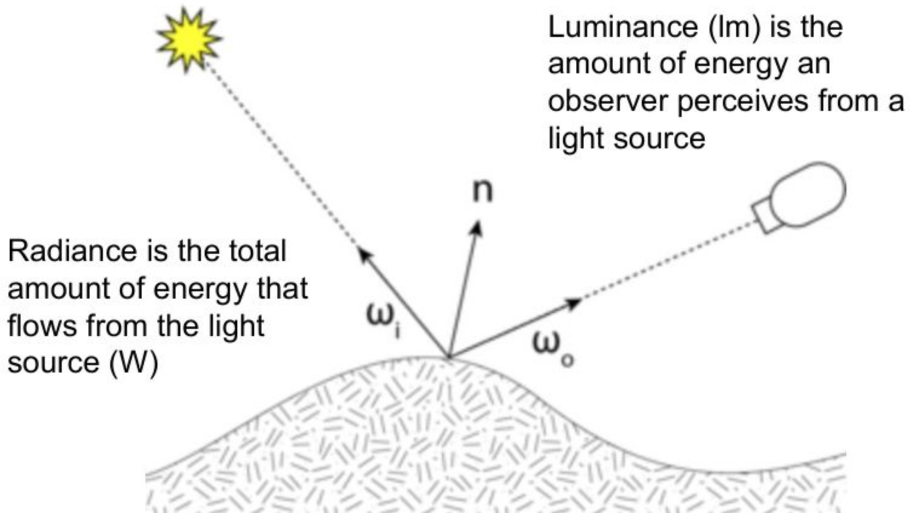
- The colors that humans perceive in an object are determined by the nature of the light reflected from the object.
    - A white object reflects light uniformly among visible wavelengths
    - A green object reflect light primarily in the range 500-570 nm range.

- **Quantities that are usually used to describe a light source**
    - **Radiance**
    - **Luminance**
    - **Brightness**
  - These values describe physical light energy and how humans perceive it.

Light Quantities

- **Radiance** - amount of light energy emitted or reflected from a surface in a specific direction.
    - "**How much light leaves a surface**"
    - Physical measurement of light

- Luminance - **amount of light perceived by the human eye from a surface**



Eg: Two lights may emit equal energy,
but the human eye may see one as brighter
depending on wavelength.

Brightness

- Brightness is a subjective descriptor of light "intensity" and one of the key factors in describing color sensation.
- It **depends on how the human brain interprets luminance**

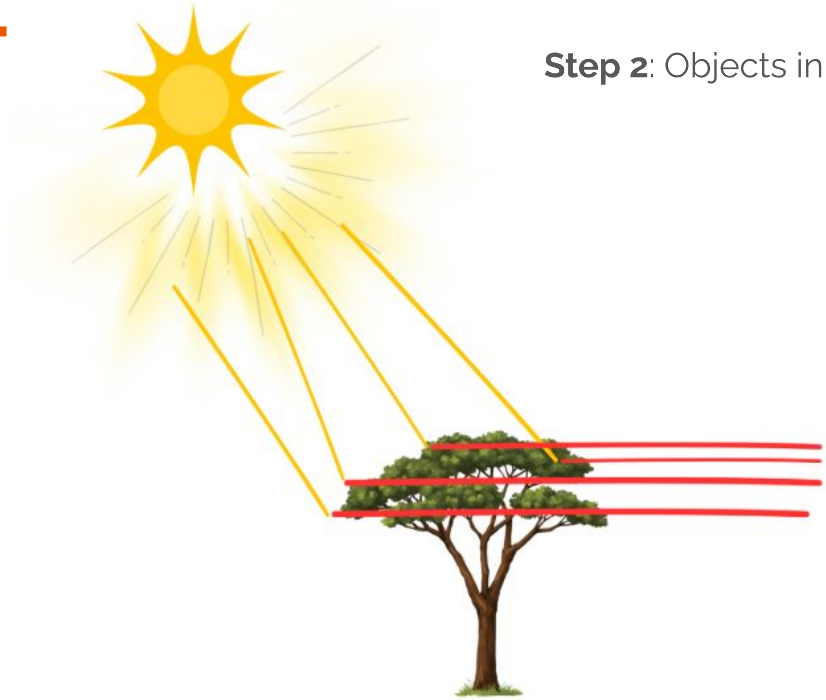
- Two areas may have the same luminance, but **one may appear brighter due to contrast with surrounding regions.**

- **Luminance of the inner square is exactly same. Perceived brightness is different.**

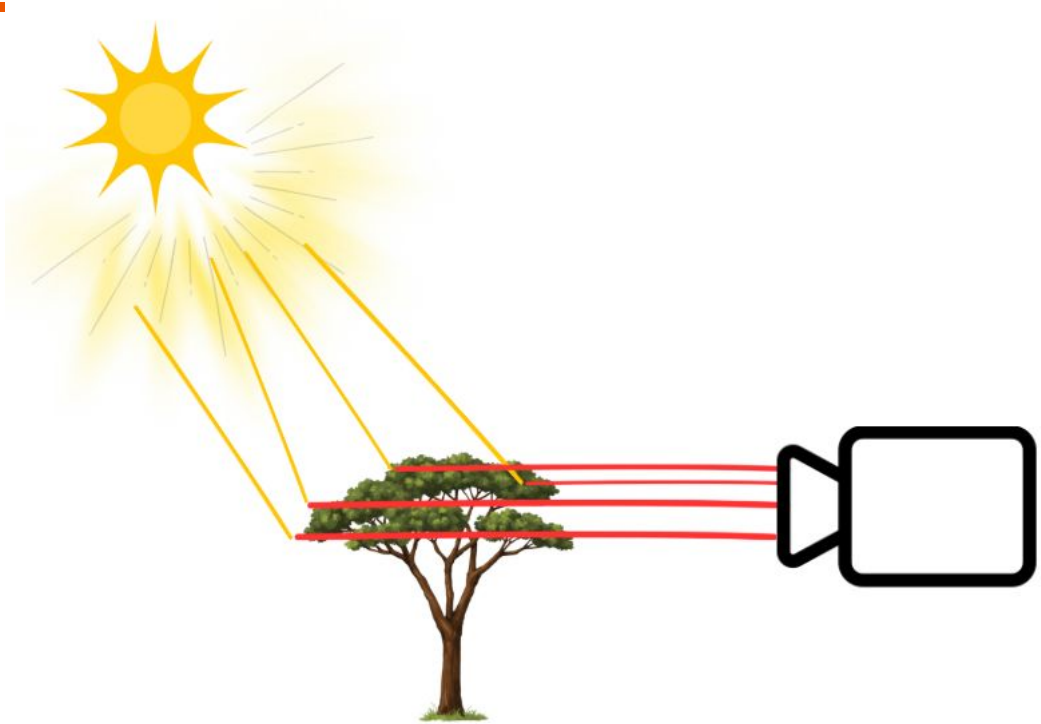
Activity: Answer the mcq questions upto Question number 28

URL: https://shorturl.at/dzxVs

Step 2: Objects in the scene reflect light



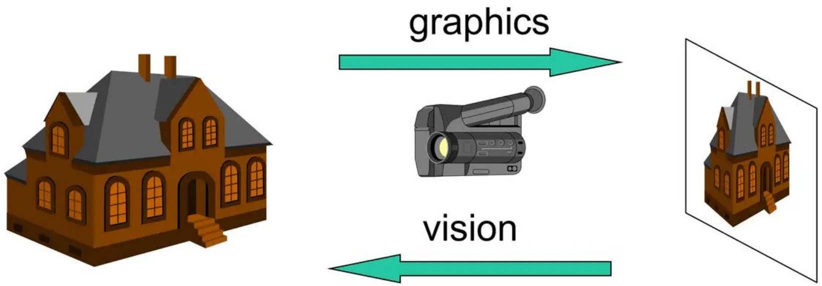
Step 3: Light travels the camera



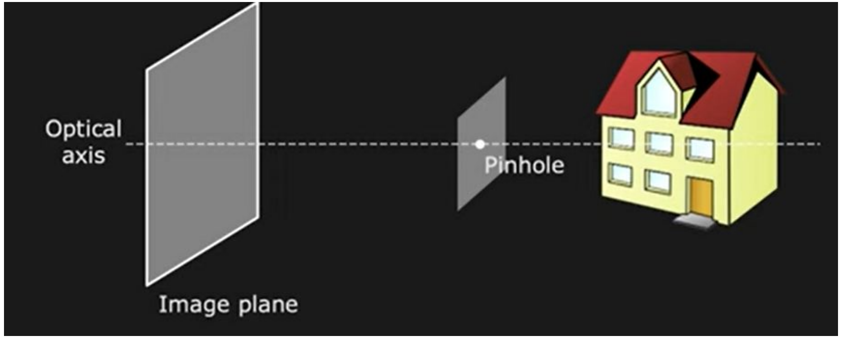
Capturing Images

- Projection of 3D scene onto 2D plane. We need to understand the geometric and photometric relation between the scene and its image.

- Pinhole Cameras - perspective projection
- Lenses
- Digital Cameras

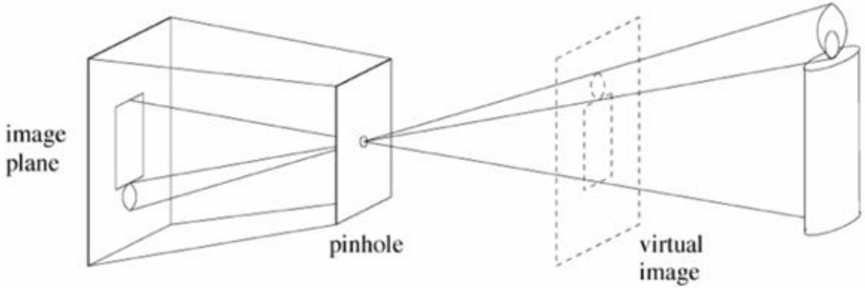


Pinhole Camera

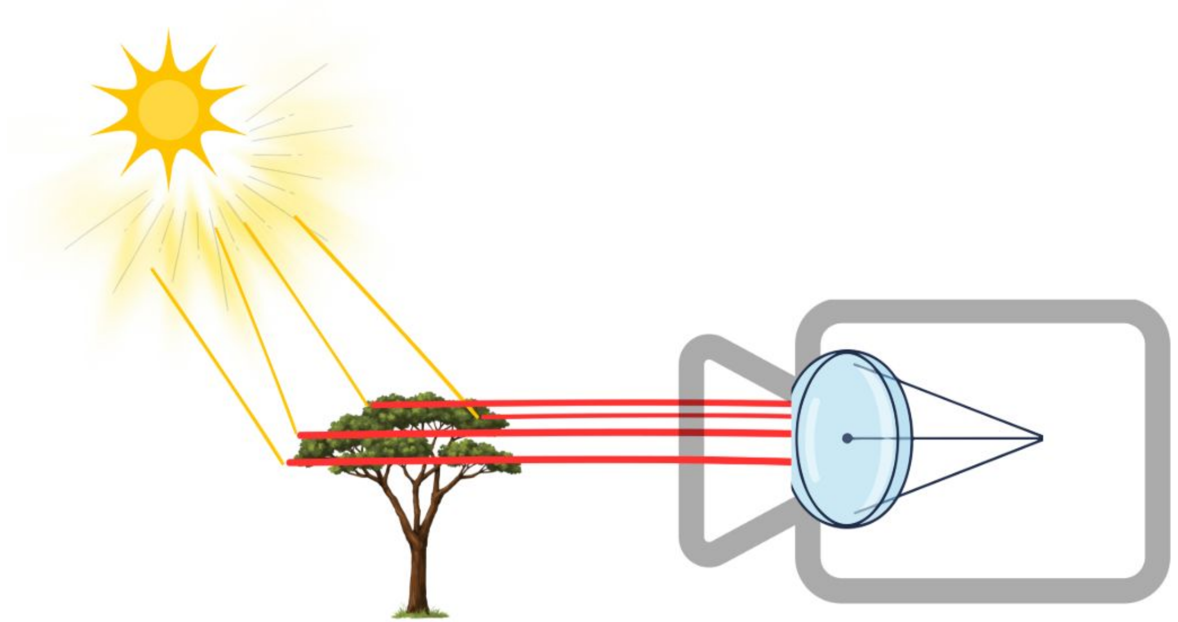


- Pinhole camera is better with clarity of the image generated but it does not gather enough light
    - So image becomes too dark
- Low light sensitivity
- Blurry images
- Long exposure times
- No focus control
- Edge distortion

- To solve this issue we use lenses
    - In modern we use converging lenses.

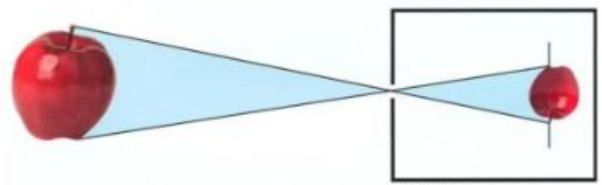


Step 4: Camera lens focuses the incoming light on Sensor plane

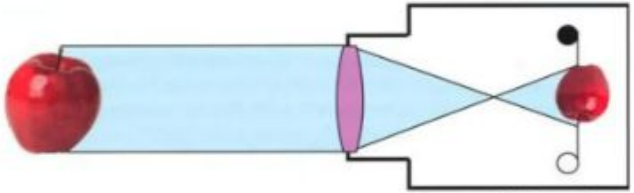


Introducing Lens

- Smaller the pinhole sharper the images but also darker.
- Larger the pinhole brighter the image but also more blurry.
- Most cameras use a converging lens to **allow more light** to enter the device.
- Zoom lenses found in cameras utilize a combination of convex and concave lenses to produce different types of images.



Pinhole Camera

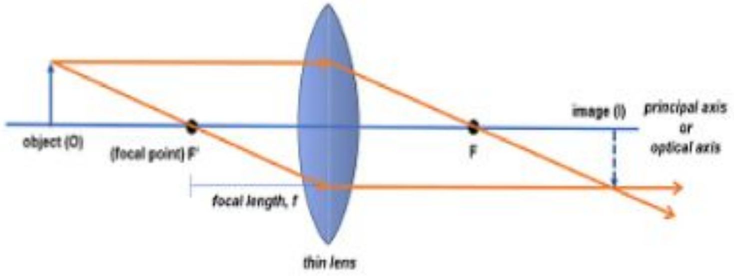


Camera with lens to focus image.

Thin lens Phenomena

- The thin lens equation defines the relationship between the focal length of a lens the distance of an object from that lens and the distance of the image formed by the lens.

Focal Length	Positive for converging lens Negative for diverging lens
Object Distance	Positive for a real object Negative for a virtual object
Image Distance	Positive for a real image Negative for a virtual image

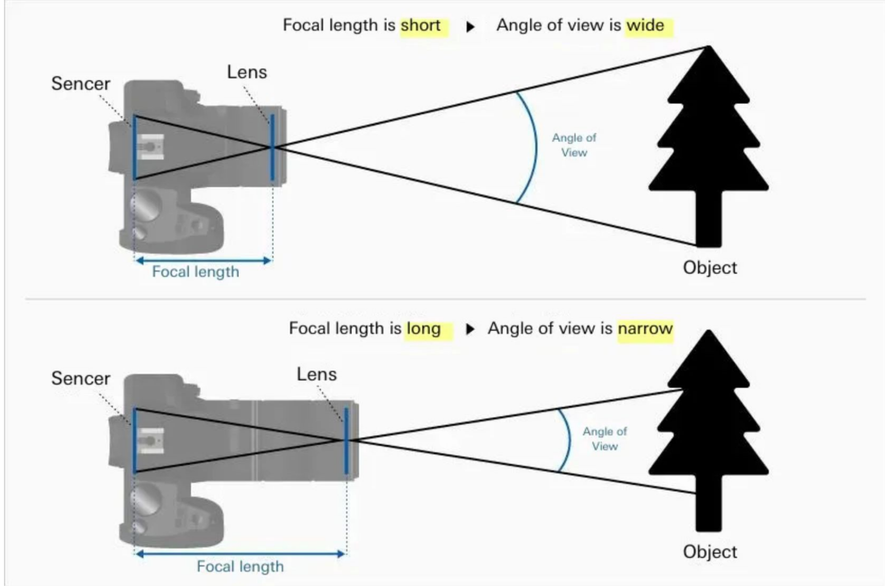


$$\frac{1}{o} + \frac{1}{i} = \frac{1}{f}$$

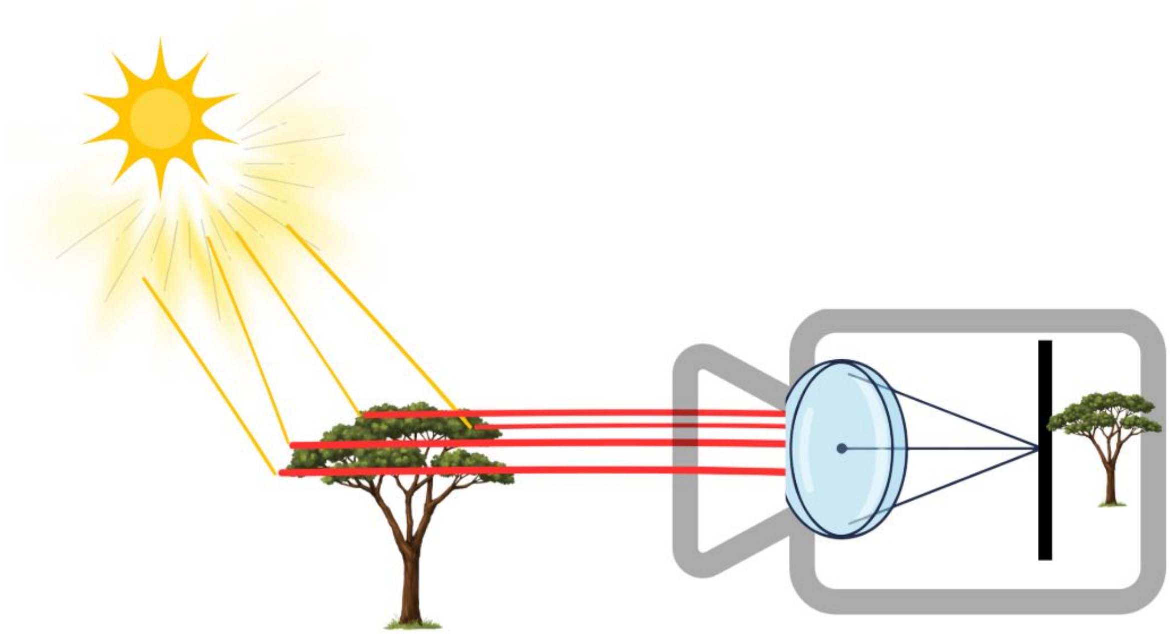
$$\frac{1}{\text{object distance}} + \frac{1}{\text{image distance}} = \frac{1}{\text{focal length}}$$

Focal length

- Focal length usually represented in millimeters (mm)
- Lens focal length tells us the angel of view - how much of the scene will be captured, how large individual elements will be
- The longer the focal length, the narrower the angle of view and higher the magnification
- The shorter the focal length, the wider the angle of view and the lower the magnification.



Step 5: 3D scene projected into 2D plane



Step 6: Sensor capture converts light energy Into electrical signals

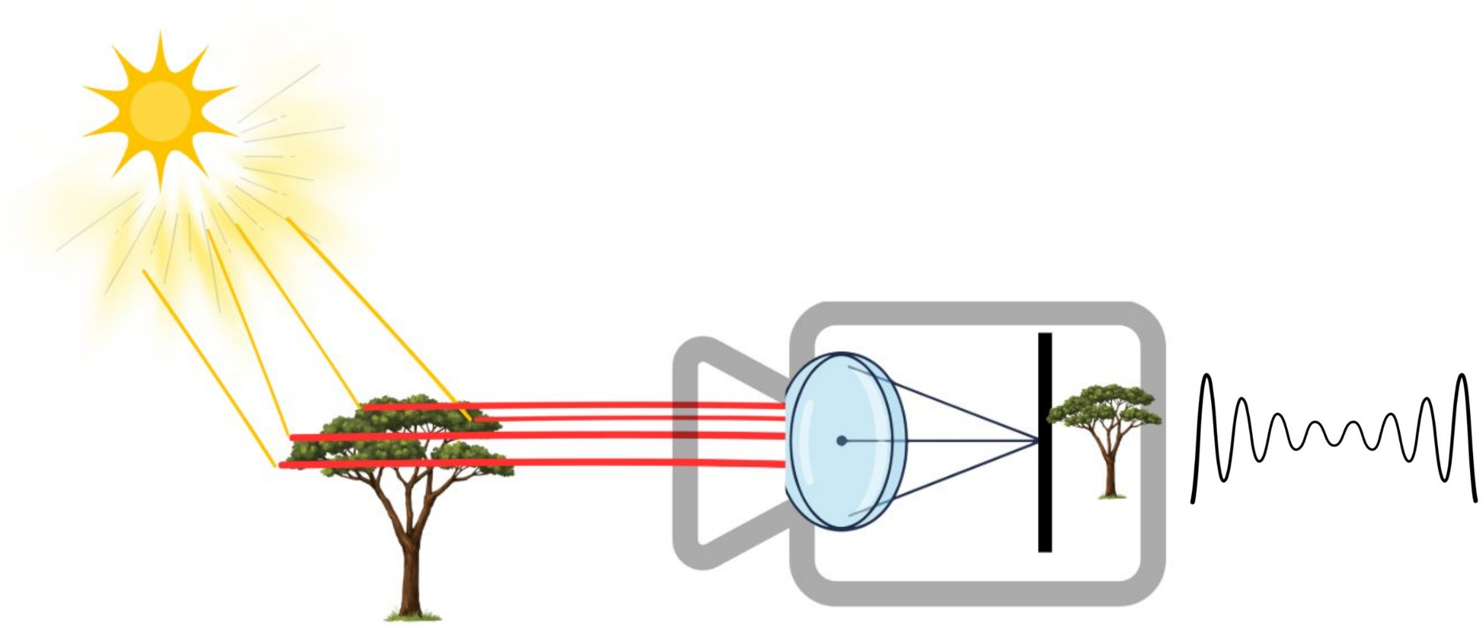
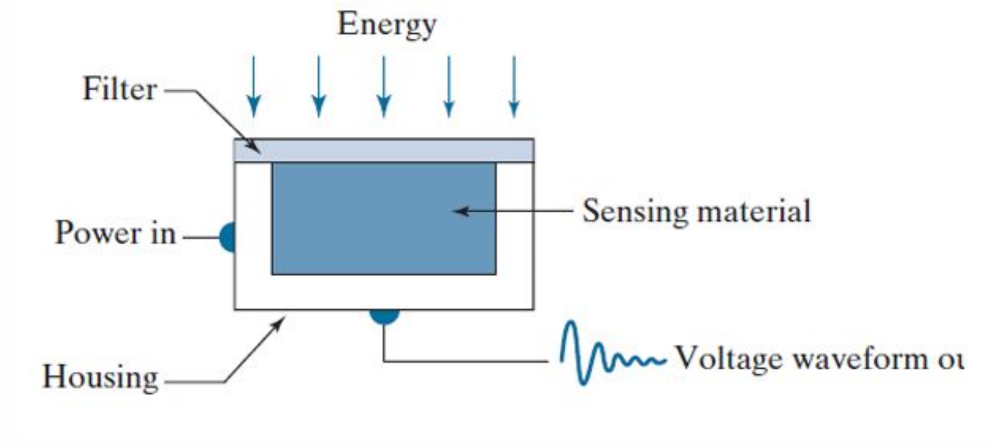


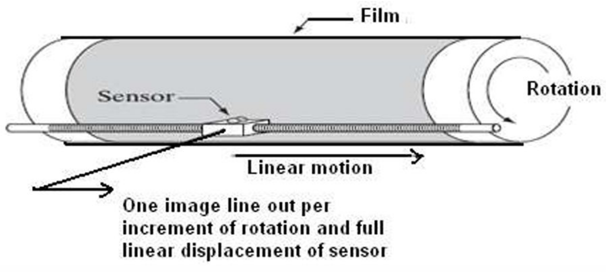
Image Sensing

- Transforming illumination energy into digital images
    - Incoming energy transformed into a voltage
    - The output voltage waveform is the response of the sensor(s)
        - A digital quality is obtained from each sensor by digitizing its response
- Three main sensor arrangements
    - Single sensor
    - Sensor strips
    - Sensor arrays

Image Sensing

- Single sensor





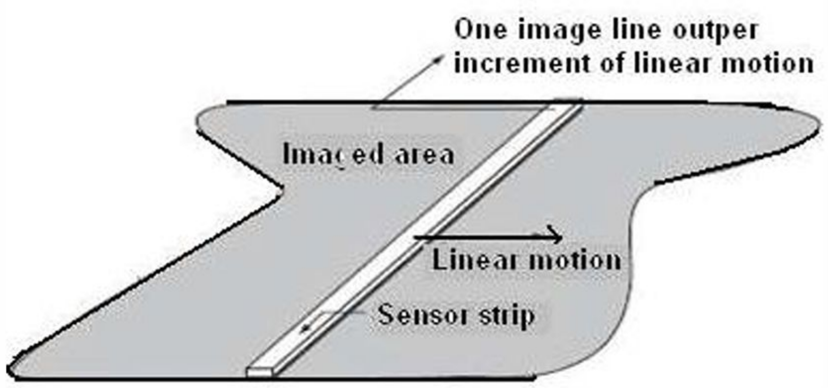
- Can we generate a 2D image using a single Sensor???
  - Need relative displacements on both x and y directions

Image Sensing

- The sensor is mounted on a lead screw
  - Motion in the perpendicular direction
- A light source is contained inside the drum
  - As the light passes through the film, its intensity is modified by the film density before it is captured by the sensor
- This “modulation” of the light intensity causes corresponding variations in the sensor voltage
  - Converted to image intensity levels by digitization.

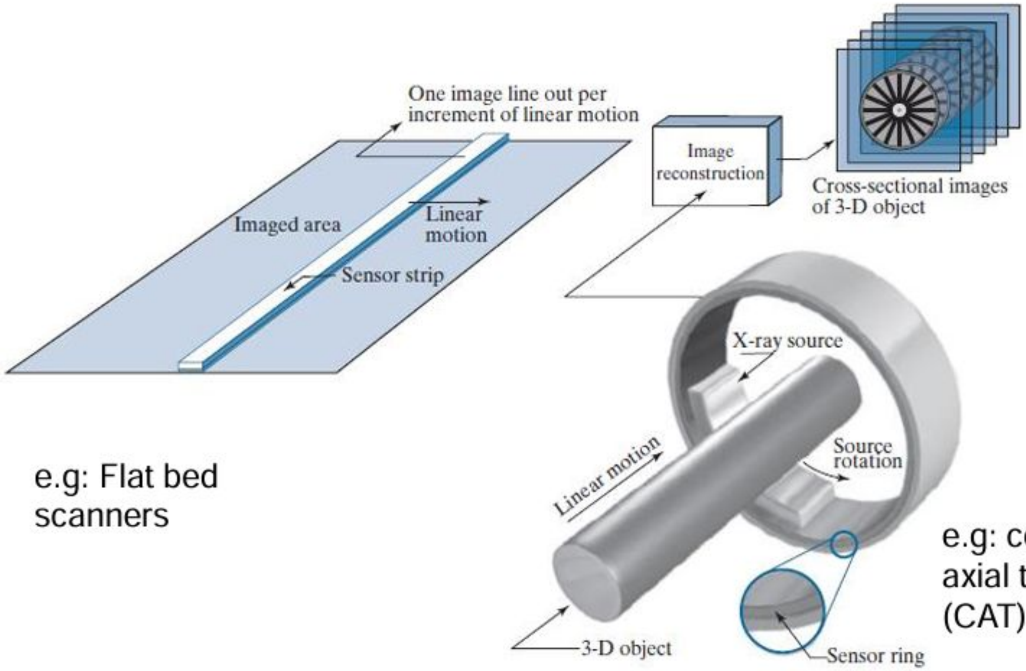
Image Sensing

- Sensor strips



- Sensor strip provides imaging elements in one direction. Motion perpendicular
- to the strips provides imaging in the other direction

Image acquisition using sensor strips

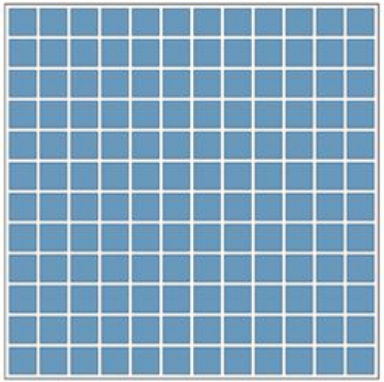


e.g: Flat bed scanners

e.g: computerized axial tomography (CAT) imaging

Image Sensing

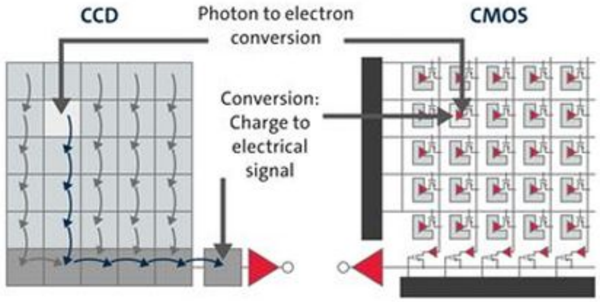
- Sensor arrays



- Complete image is obtained by focusing the energy pattern onto the array
- Motion not necessary

Capturing Digital Images

- Light falling on an image sensor is usually picked u| by an active sensing area.
- Charge-coupled Device (CCD) - sensitive photon detector
- Complementary Metal Oxide on Silicon (CMOS)
- CCDs are prone to “Blooming”
- CCD sensors outperformed CMOS in quality-sensitive applications



Two different principles: CCD vs CMOS

A/D conversion from voltage to a digital signal can happen in 2 ways: At the end of each row/column (CCD) or directly at each sensing cell (CMOS).

As a rule of thumb C**MOS is in general better** but CCD is faster and works better in low light

Step 7: Digitization

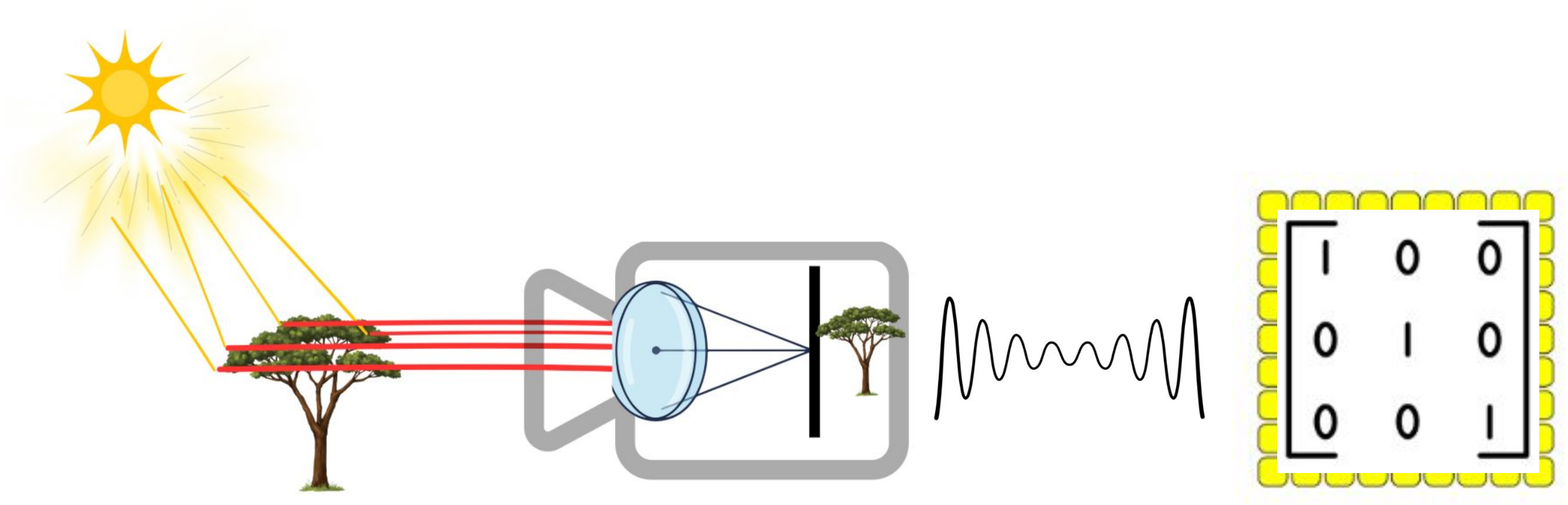
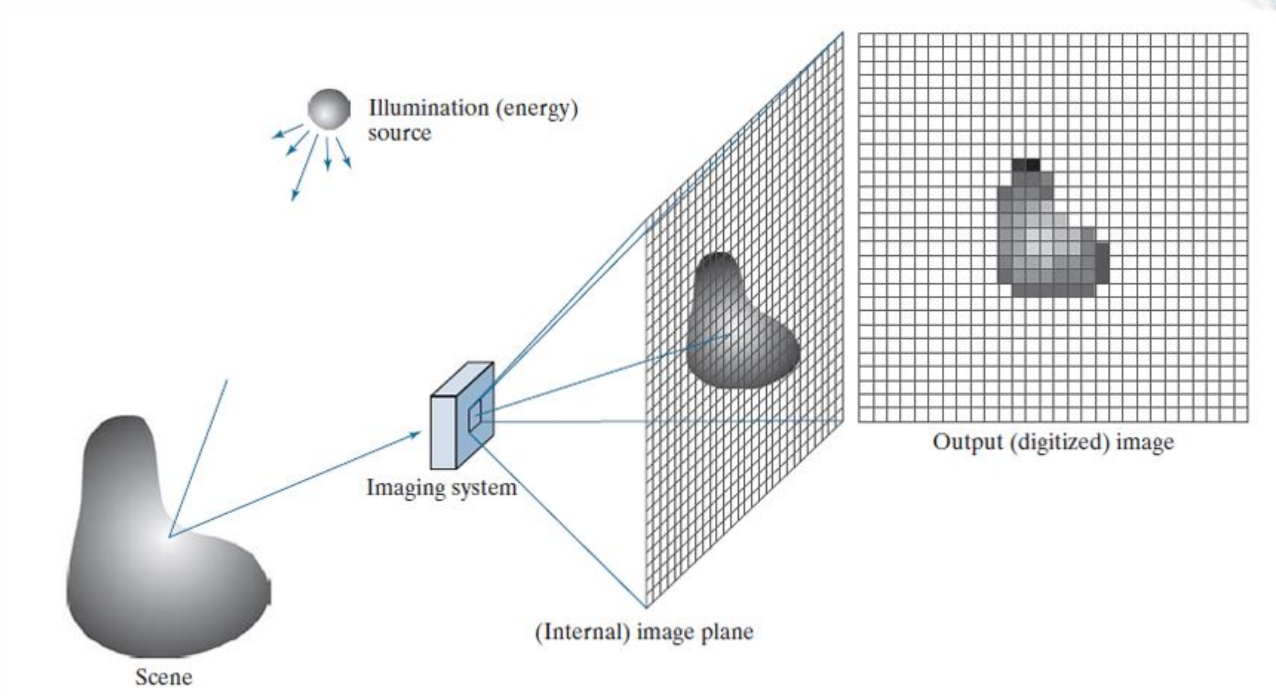


Image formation using sensor arrays (process)

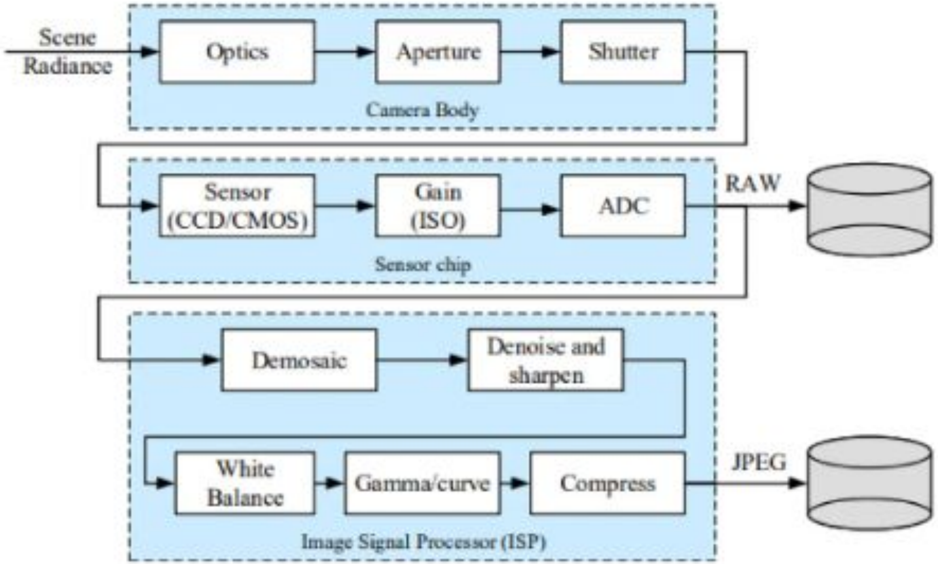


Activity: Answer the questions upto 40

Digital Cameras

- Image sensing pipeline
- Various sources of noise
- Typical digital post processing steps

ISO - Make low light signal stronger(amplification)
ADC = Analog to digital conversion
Demosaicing - color construction (Bayer color filter
    Each pixel capture R, G, B
White balance - corrects color based on lighting
Noise reduction - removes sensor noise
Gamma/tone curve - adjust how brightness value
Are mapped so image looks natural on display.



Assignment : Graded

Image formation

- Image formation is an analog to digital conversion of an image with the help of 2D sampling and quantization techniques that is done by the capturing devices like cameras.

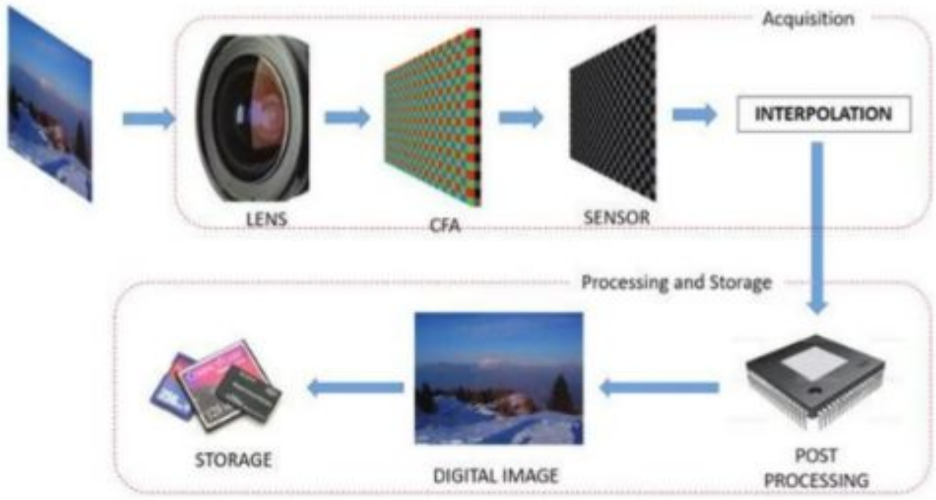


Image formation

- An image can be modeled as a function,

$$I : \Omega \subset \mathbb{R}^2 \rightarrow \mathbb{R}$$

The domain is a (usually rectangular) subset of the real image plane.

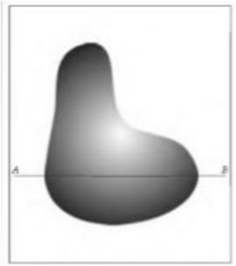
$$I(x, y)$$

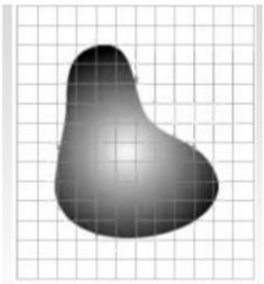
is proportional to the amount of light energy that is collected at the image plane at coordinated x, y. The amount of energy is called intensity.

- A grayscale image
  - Values in the interval [0, L-1] Dynamic range
    - 0 - black
    - L-1 - white
    - Intermediate values - shades of gray
- Contrast - Difference in intensity between the highest and the lowest values
  - High dynamic range => high contrast

Image sampling and quantization

- Output of most sensors - a continuous voltage waveform
  - Need to convert to digital form
- When an image is digitized it is sampled in
  - SPACE (x, y) : image sampling
    - Digitizing the coordinate values
  - AMPLITUDE f(x,y) : gray level quantization
    - Digitizing the amplitude





In sampling we consider an equispaced grid of values in a given area (matrix). Each sample is called a pixel.

Image sampling and quantization

- The intensity (output) of the function must also be discretized in a finite set of values to be digitized and used in a computer.
- Image codomain is divided into a set of values and each f(x, y) is rounded to closed one.

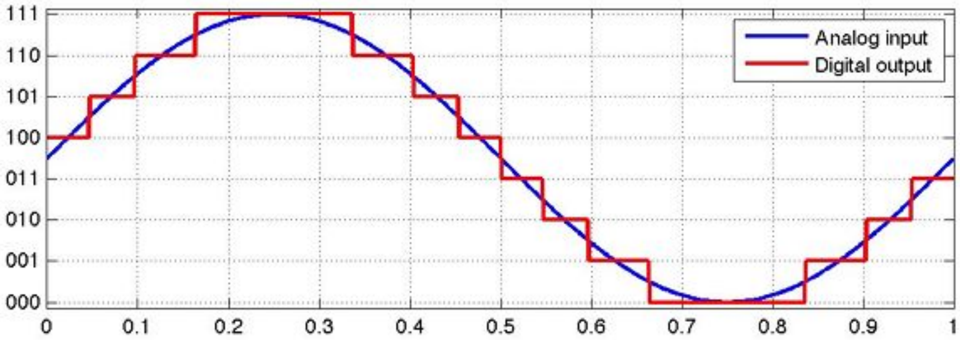


Image sampling and quantization

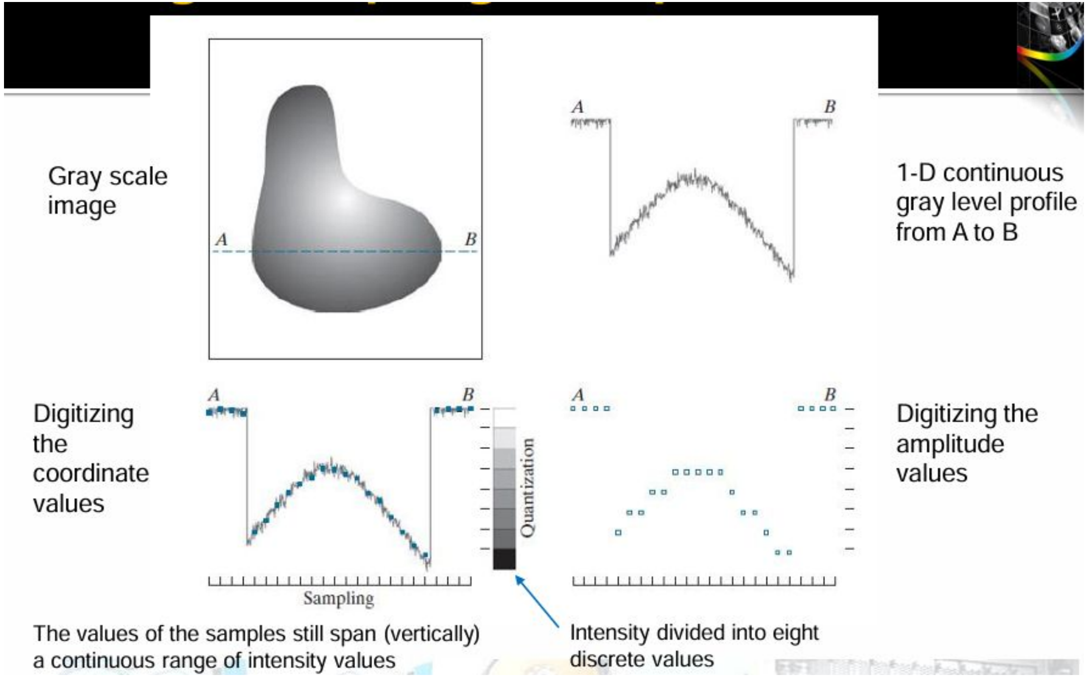
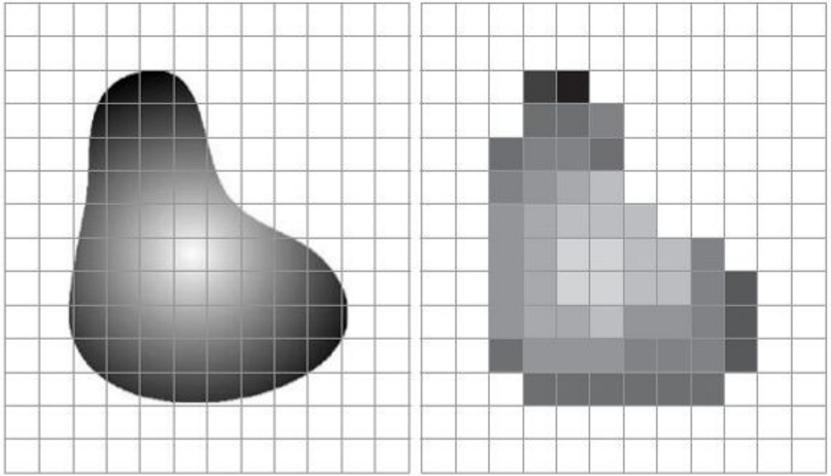


Image sampling and quantization

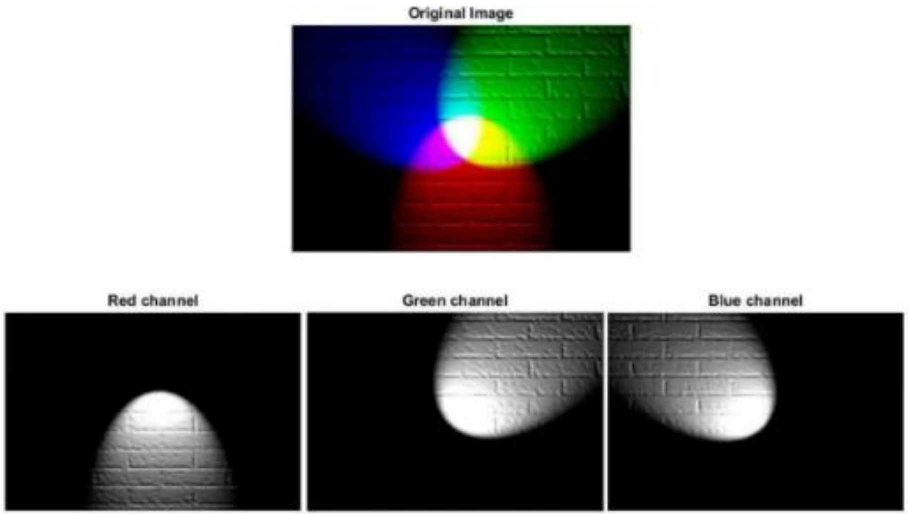


Continuous image projected onto a sensor array

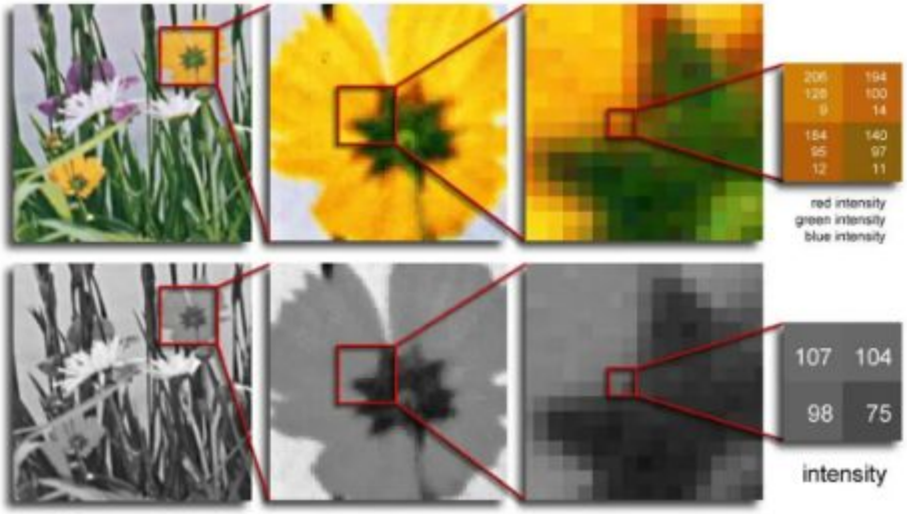
Result of image sampling and quantization

Image representation

- RGB image with r, g , b components



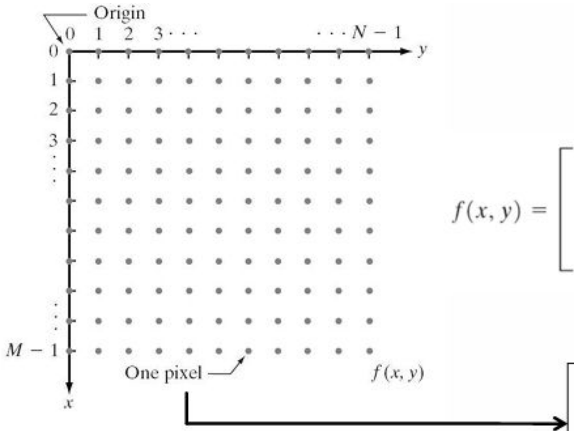
RGB vs Grayscale images



Binary vs GrayScale vs RGB images

Representing digital Images

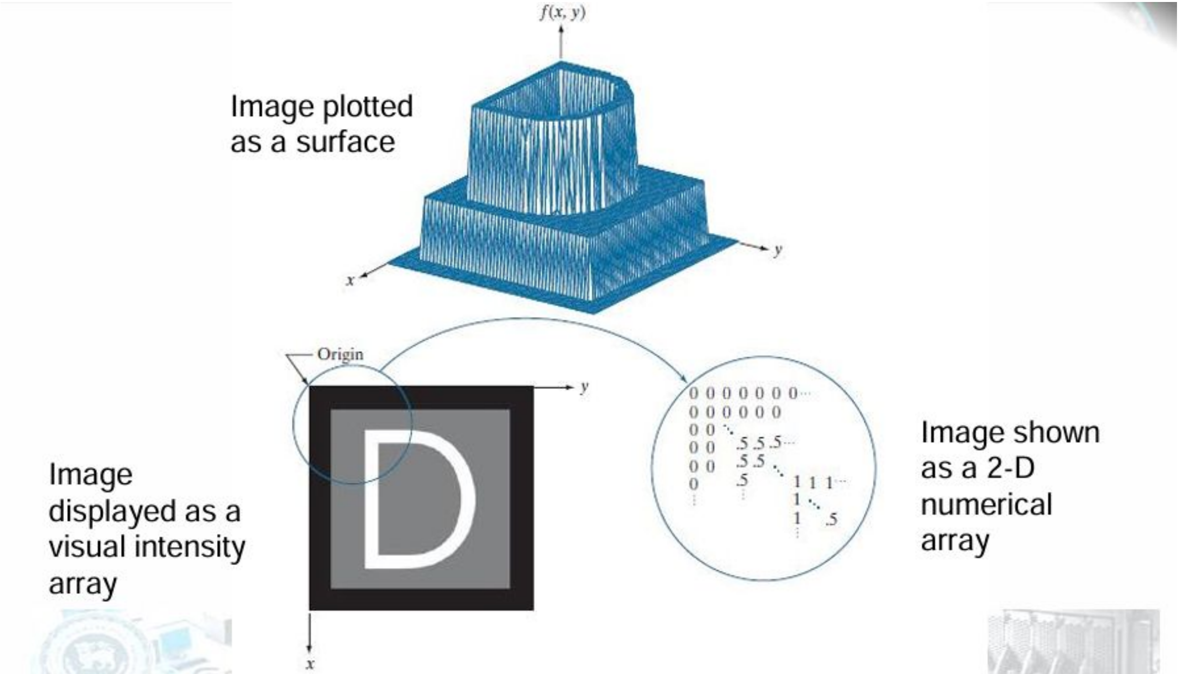
- Sampling and quantization -> matrix of real numbers



$$f(x,y)=\begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

2D – pixel (picture element)
3D – voxel (volume element)

Representing digital Images

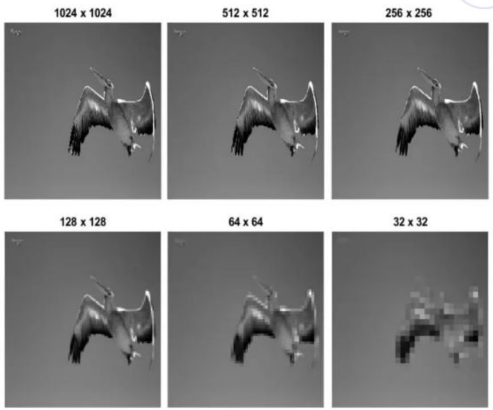


Representing digital Images

- Intensity and Matrix representation are the most common to be used in practice.
- The digitization process requires that decisions be made regarding the matrix size M, N and the number L (usually a power of 2) of quantized intensity levels
- M and N are related to the spatial resolution of an image
- L is the related to the intensity resolution

Spatial resolution

- The spatial resolution of an image is determined by how sampling was carried out
- Spatial resolution is a measure of the smallest discernible detail in an image.
- Usually defined as dots (pixels) per unit distance.
- Note: to be meaningful, measures of spatial resolution must be stated with respect to spatial units
- A 1024 x 1024 pixel image is not meaningful without stating the spatial dimensions encompassed by the image (the size of each pixel in the 3D world)



- 3 measures which we see often relating to Image Size/ Resolution
  - Pixel count eg: 3000x2000 pixels
  - Physical count eg: 8'' x 10''
  - Resolution eg: 240 pixels per inch(PPI)

Spatial resolution

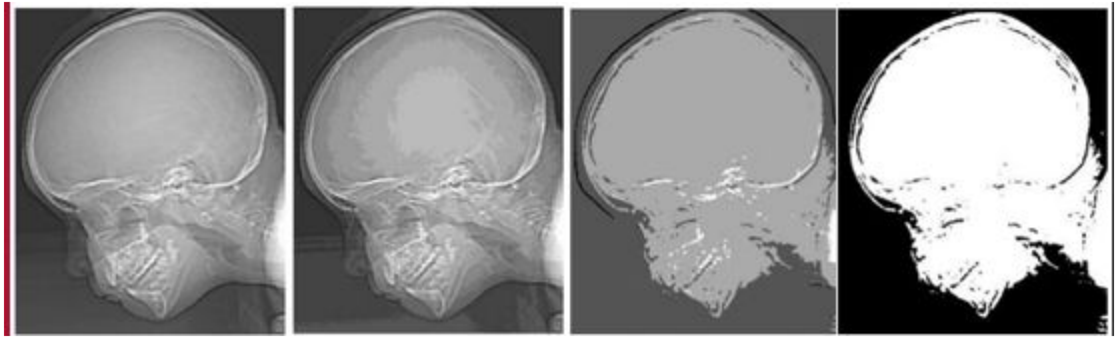
1250 dpi

300 dpi

72 dpi

Intensity resolution

- Intensity resolution refers to the smallest discernible change in the intensity level.
- Usually it is a power of 2 (8 bits most common)



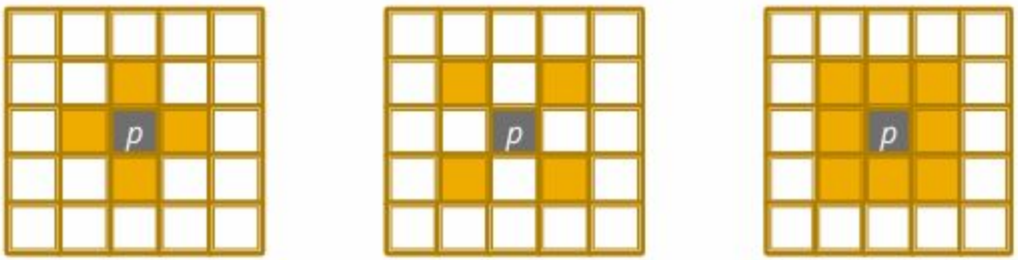
256 levels 16 levels 4 levels 2 levels

Image interpolation

- Used in zooming, shrinking, rotating and geometrically correcting digital images.
    - Nearest neighbour interpolation
    - Bilinear interpolation
    - Bicubic interpolation

Relationships between pixels

- Neighbors of a Pixel



$N_4(p)$, 4-neighbors (edge neighbors)

$N_D(p)$, diagonal neighbors

$N_8(p)$, 8- neighbors (4- neighbors + diagonal neighbors)

Adjacency, Connectivity

- Two pixels are adjacent if they are neighbors and their intensity levels V satisfy some criteria of similarity
  - V- set of gray-level values
    - For a binary image, V={1}

Adjacency, Connectivity

- 4–adjacency
  - twopixels p and q with values from V are 4-adjacent if q is in the set N4(p)

- 8-adjacency
  - Twopixels p and q with values from V are 8-adjacent if q is in the set N8(p)

Adjacency, Connectivity

- m-adjacency-(mixed adjacency).
  - Two pixels p and q with values from V are m adjacent if
    - q is in $N_4(p)$, or
    - q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V

Question

		S_1				S_2			
0	0	0	0	0	0	0	1	1	0
1	0	0	1	0	0	1	0	0	1
1	0	0	1	0	1	1	0	0	0
0	0	1	1	1	0	0	0	0	0
0	0	1	1	1	0	0	1	1	1

Question (cntd.)

For V={1}, determine whether these two subsets are

(a) 4-adjacent,

(b) 8-adjacent, or

(c) m-adjacent

Answer

		S_1					S_2		
0	0	0	0	0	0	0	1	1	0
1	0	0	1	0	0	1	0	0	1
1	0	0	1	0	1	1	0	0	0
0	0	1	1	1	0	0	0	0	0
0	0	1	1	1	0	0	1	1	1

Answer (cntd.)

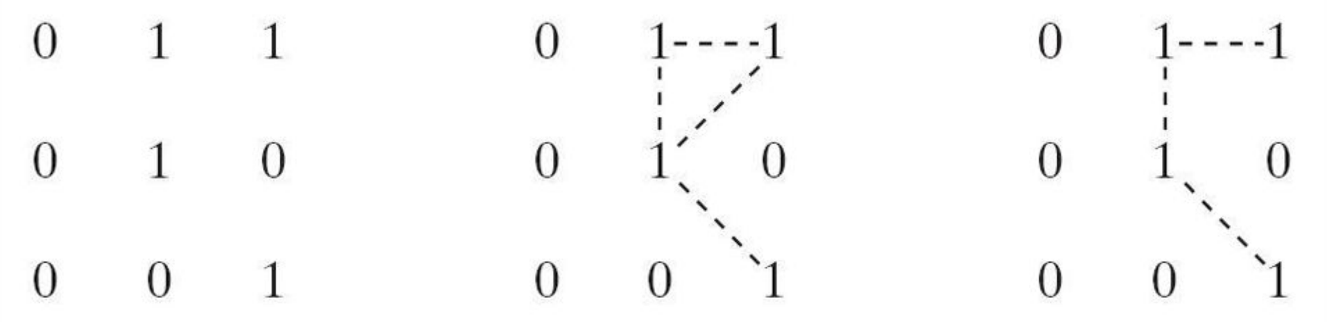
(a)S1 and S2 are not 4-connected because q is not in the set N4(p)

(b)S1 and S2 are 8-connected because q is in the set N8(p)

(c)S1 and S2 are m-connected because (i) q is in ND(p), and (ii) the set N4(p) $\cap$ N4(q) is empty

Why m-adjacency?

- To eliminate ambiguities in 8-adjacency



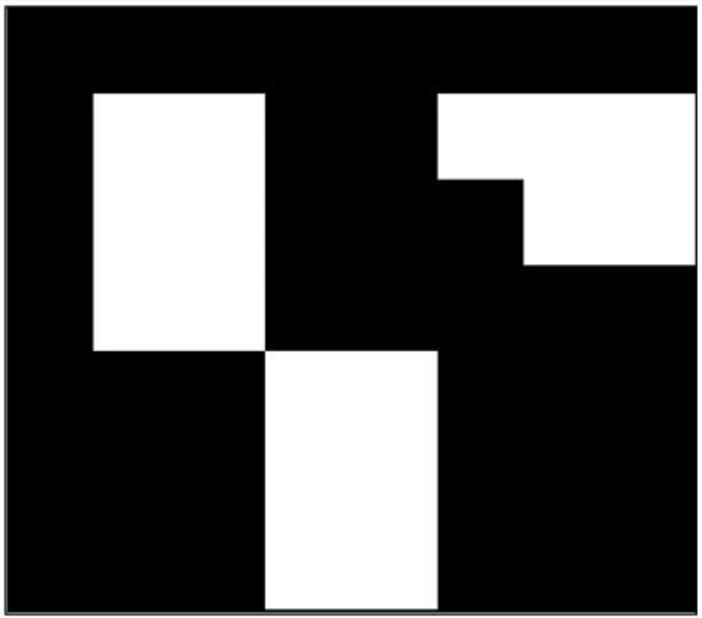
Why m-adjacency? (cntd.)

- to connect between two pixels (finding a path between two pixels):
    - In 8-adjacency -multiple paths between two pixels
    - in m-adjacency-only one path between two pixels

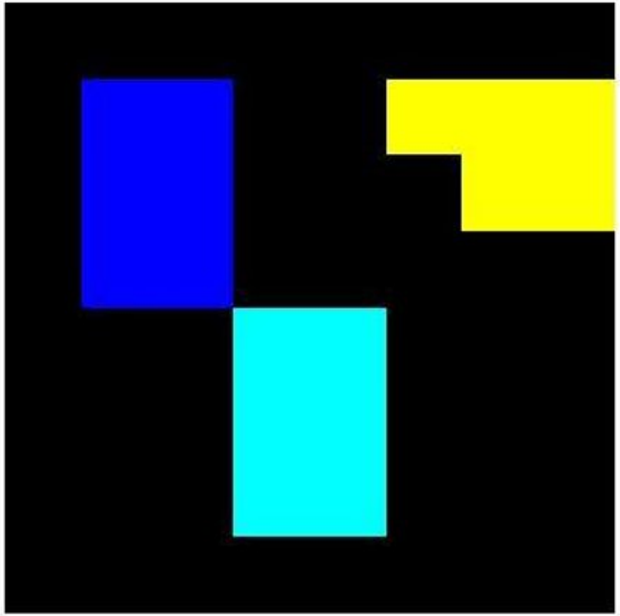
Adjacency, Connectivity

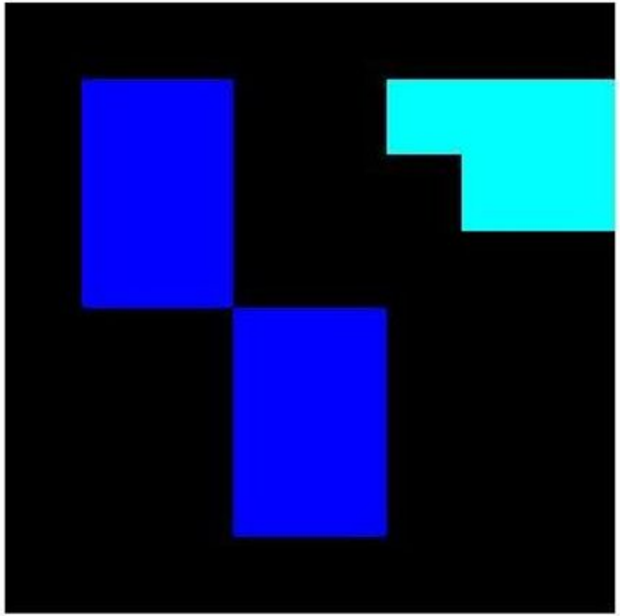
- Connectivity
- Let S represent a subset of pixels in an image
  - Two pixels p and q are said to be connected in S if there exists a path between them consisting entirely of pixels in S
- For any pixel p in S, the set of pixels that are connected to it in S is called a connected component of S

How many objects?



How many objects? (cntd.)





Paths

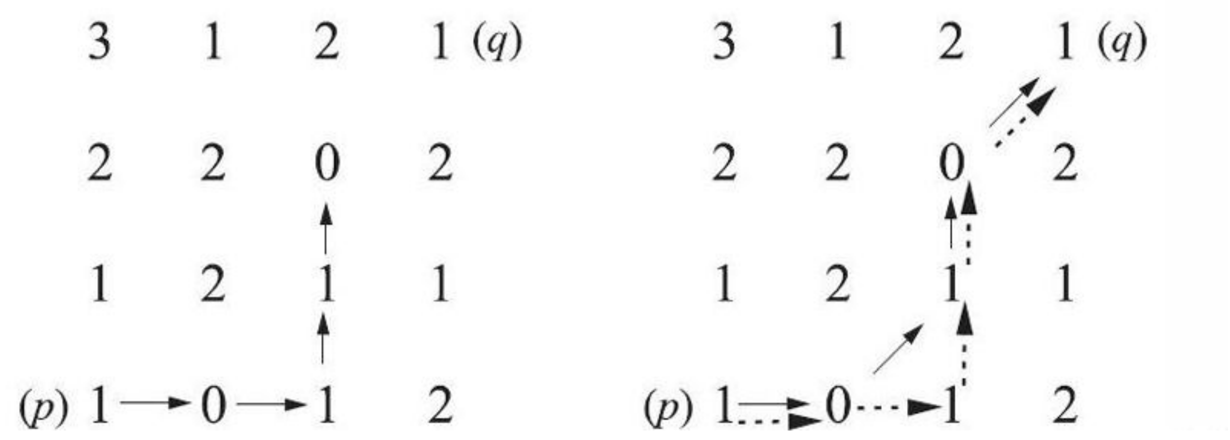
Let V={0, 1}. Compute the lengths of the shortest 4-, 8-, and m-path between p and q. If a particular path does not exist between these two points, explain why.

$$
\begin{array}{rcccl}
 & 3 & 1 & 2 & 1\,(q) \\
 & 2 & 2 & 0 & 2 \\
 & 1 & 2 & 1 & 1 \\
(p) & 1 & 0 & 1 & 2
\end{array}
$$

Paths (cntd.)

When V = {0,1}, 4-path does not exist between p and q.

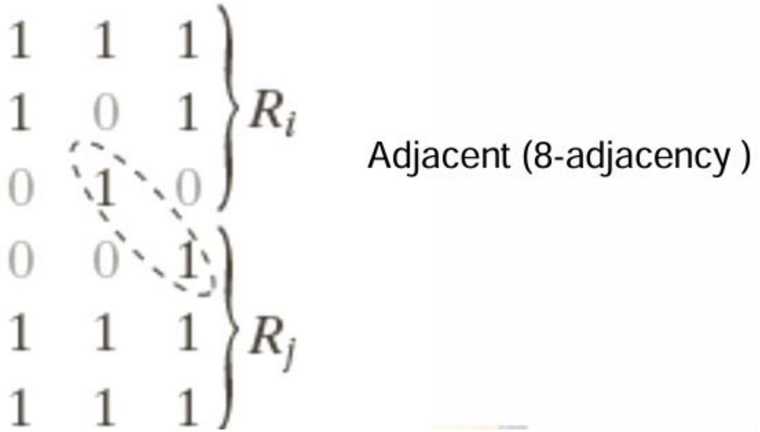
- It is impossible to get from p to q by traveling along points that are both 4-adjacent and also have values from V



Adjacent and disjoint regions

Let R be a subset of pixels

- Two regions Ri and Ri are adjacent if their union forms a connected set
- Regions that are not adjacent are called disjoint



Region and boundary

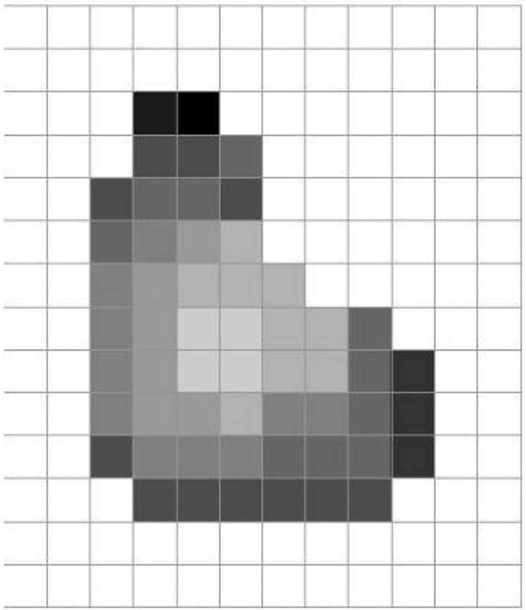
- Let R be a subset of pixels in an image. We call R a region of the image if R is a connected set.

- The boundary(also called border or contour) of a region R is the set of pixels in the region that have one or more neighbors that are not in R.

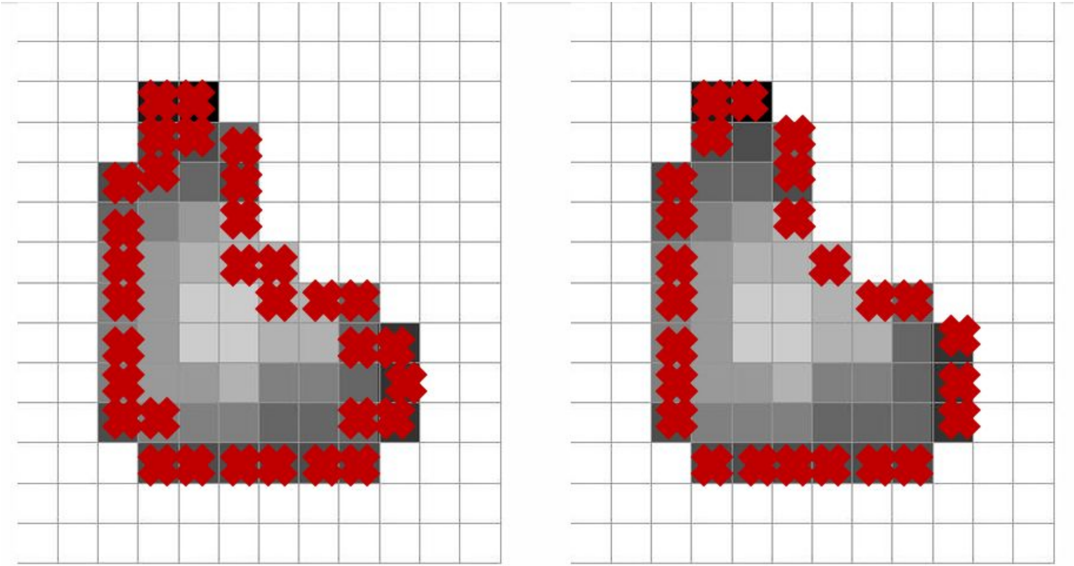
Boundary of an object

Set of pixels in a region that has at least one background neighbor

- Need to define the connectivity that is used



Boundary of an object (cntd.)



4-connected boundary 8-connected boundary

Distance Measures

For pixels p, q, and z, with coordinates (x, y), (s, t), and (v, w), respectively, D is a distance function or metric if

- D(p, q) 0 , (D(p, q)=0 iff p=q)
- D(p, q)=D(q, p)
- D(p, z) D(p, q)+D(q, z)

Distance Measures (cntd.)

- Euclidian distance

$$D_e(p, q) = [(x - s)^2 + (y - t)^2]^{1/2}$$

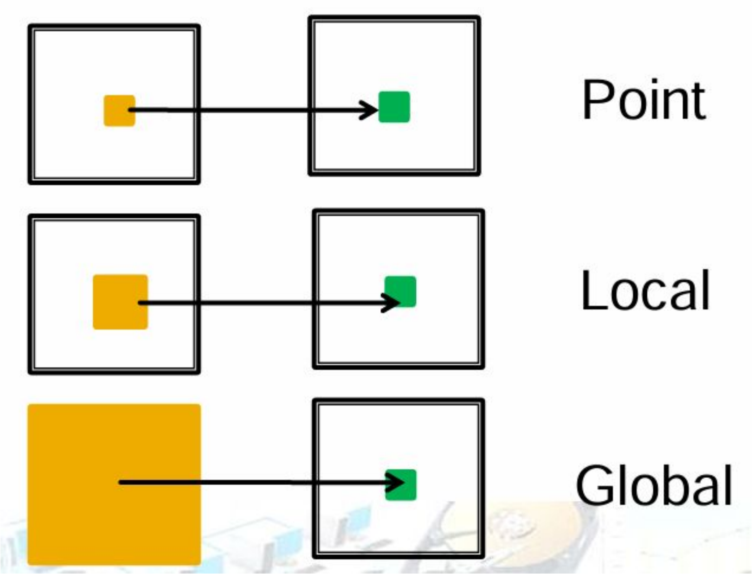
▪ the pixels having a distance less than or equal to some value r from (x, y) are the points contained in a disk of radius r centered at (x, y)) 2 + (y − t) 2] 1 / 2

Distance Measures (cntd.)

- City-block distance (D4 distance)
- Chessboard distance (D8 distance)

Image Operations

Types of operations carried out on digital images can be categorized into 3 categories



Thank you.